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MILITARY SPECIFICATION

CUVETTE, SPECTROPHOTOMETER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE AND CLASSIFICATION

* 1.1 Scope. This specification covers replacement cuvettes for use with a specific photometric spectrophotometer (see 6.1).

1.2 Classification. Cuvettes shall be furnished in the following sizes, as specified, for the appropriate stock number and item identification.

Size A - 10 x 75 millimeters (mm.)
Size B - 19 x 150 mm.

2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Fiberboard.
PPP-B-676 - Boxes, Set-Up, Paperboard.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, Code ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6640

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MILITARY

MIL-L-10547 - Liners, Case and Sheet, Overwrap; Water-Vaporproof, or Waterproof, Flexible.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM Designation C148 - Polariscopic Examination of Glass Containers.
ASTM Designation C162 - Standard Definitions of Terms Relating to Glass and Glass Containers.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

3. REQUIREMENTS

* 3.1 Material. Cuvettes shall be made of clear, transparent glass and shall be free from cracks, chips, breaks, crizzles, checks, mars, blemishes, stones, scratches, pits, seeds, bubbles, striae, blisters, and all color defects.

3.1.1 Annealing. Ring strain shall be not greater than that of temper No. 1, when tested as specified in 4.4.1. Any longitudinal strain, if present, shall be faint and highly diffused.

3.2 Construction. Cuvettes shall be made of clear, colorless glass (see 3.1.1), and shall be symmetrically tubular (round) in shape, with one end having a rounded closure. The open end shall be lipless, square cut 90 degrees to the vertical axis, and fire-polished.

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3.3 Dimensions.

* 3.3.1 Diameter and length. External diameter and length of the cuvettes shall be as listed below and shall be designated as "B" cuvettes (see 3.4).

Size	Outside diameter (inch)	Length (mm.)
A	0.365 to 0.380	75 ± 2
B	0.720 to 0.740	150 ± 3

* 3.4 Optical transmission. The optical transmission of size A and B cuvettes shall be equal to a standard "B" cuvette reading ± 0.6% transmittance, when tested as specified in 4.4.2. Cuvettes shall afford the specified transmission in all positions of rotation with respect to the major axis of the tube. Cuvettes not designed to be usable in all positions shall be indexed to indicate correct positioning of the tube. These indexed cuvettes shall be within the specified transmission when rotated within ± 15 degrees from the indexed position.

* 3.5 Identification marking. Cuvettes shall be permanently and legibly marked with a "B" near the open end, the name or registered trademark of the manufacturer, and, when applicable, marking designating the optical index of the cuvette.

* 3.6 Workmanship. Workmanship shall be first class throughout. Cuvettes shall be free from defects which detract from their appearance or impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification to assure that supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

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4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and table I.

TABLE I. Sampling for examination.

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	1.5
Minor defects	II	4.0
For dimensional examination	S-2	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table II.

TABLE II. Sampling for tests.

Component (unit of product)	Characteristic	Requirement	Test procedure	Inspection level	AQL
End item	Strain	3.1.1	4.4.1	S-3	*
	Optical transmission	3.4	4.4.2	S-3	*

* Acceptance number shall be zero.

4.3 Examination. The tubes shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the classification of defects listed in table III.

TABLE III. Classification of defects.

Categories	Defects*
Major	
101	Item identification missing, illegible or incomplete.
102	Cuvette not free of cracks, chips or breaks.
103	Cuvette not free of crizzles or checks.
104	Mouth edge of cuvette not cut at right angles to vertical axis of tube.
105	Cuvette not clear or free of color defects.
106	Cuvette not free of sharp edges or protrusions.
107	Mouth edge of cuvette not fire-polished.
108	Cuvette not free of seeds, stones, pitting, blisters or etched marks.

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TABLE III. Classification of defects (continued).

Categories	Defects
Minor	
201	Cuvette not free of scratches or striae.
202	Mouth edge of cuvette not free of irregularity.
203	Cuvette not free of dirt or any foreign matter.
204	Cuvette not free of mars or blemishes.

* Examination shall not be restricted to the classified possible defects listed above. Definitions of glass defects shall be in accordance with ASTM Designation C162.

* 4.3.2 Dimensional examination. Cuvettes shall be examined for defects in dimensions. Any dimension not within the tolerance specified herein shall be considered as a defect.

* 4.3.3 Packaging inspection. The inspection of the packaging and packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

* 4.4.1 Strain test. Cuvettes shall be tested in accordance with ASTM Designation C148.

* 4.4.2 Optical transmission. The optical transmission of a standardized solution of nigrosine in the sample cuvettes shall be measured in a suitable spectrophotometer (see 6.1) at a wavelength of 580 millimicrons (mu.) and compared against the applicable size standard "B" cuvette. The spectrophotometer shall be set at 100% transmission (zero density) using distilled water in a standard cuvette. Standardize a solution of nigrosine so that the optical density at a wavelength of 580 mu. is 0.400.

Standardization of Nigrosine Solution may be accomplished as follows:

(a) Nigrosine concentrate. Add 1.4 gm. of nigrosine crystals to 100 ml of distilled water. Allow to soak for several hours with occasional stirring Add boiling beads and boil for 10-15 minutes. Decant or siphon solution into another flask, allowing the last 5 cc. of residue to remain. Allow to stand undisturbed until ready for use, then filter.

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(b) Measure optical density. Dilute 2 ml. of filtered concentrate to 1000 ml. Measure optical density (O.D.) at 580 mu. using the standard cuvette.

(c) Dilution. Dilute solution with distilled water so that the optical density of the final dilution is 0.400, using the following formula:

$$Lf = \frac{LM\ (O.D.M.)}{0.400}$$

Lf is final volume in liters after appropriate dilutions.
LM is volume in liters of solution made up.
O.D.M. is measured optical density.

Amount of water required for dilutions is Lf - LM.

5. PACKAGING

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Unit of issue. One box containing 12 cuvettes as specified, constitutes one unit of issue.

5.1.1.1 Level A.

5.1.1.1.1 Unit packaging. Each unit of issue shall be packaged in a box of appropriate size constructed in accordance with PPP-B-636, type CF, class domestic. Box shall be designed with suitable interior fitting or shall contain sufficient cushioning material to ensure against shifting of, or damage to the contents. Closure of box shall be in accordance with the appendix of the box specification.

5.1.1.1.2 Intermediate package. Twelve units of issue shall be packed in a box constructed in accordance with PPP-B-566, PPP-B-676, or PPP-B-636, type CF, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.1.1.2 Level C. Each unit of issue shall be packaged in a manner that will afford adequate protection against corrosion and physical damage during shipment from supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Thirty-six (36) units of issue shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather resistant, grade V3C. Closure and strapping shall be as specified in the appendix of the applicable box specification.

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5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Fiberboard box shall be waterproof as specified in the appendix of PPP-B-636.

5.2.2 Level B. Thirty-six (36) units of issue shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, overseas type; PPP-B-621, class 1; or PPP-B-636, type CF, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The packaged cuvettes shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receving activity. These packs shall conform to the applicable carriers' rules and regulations.

5.3 Marking. Each unit, intermediate package and exterior container shall be marked as specified in MIL-STD-129.

6. NOTES

* 6.1 Intended use. The specified cuvettes are intended to be used with the following instrument that is listed in the Federal Supply Catalog.

National Stock Number	Item Identification
6650-00-144-4555	SPECTROPHOTOMETER, PHOTOELECTRIC, 110 or 220 Volts, 50/60 Cycle AC Volts

* 6.2 Ordering data. The procurement document should specify the following.

(a) Title, number and date of this specification.
(b) National Stock Number (NSN).
(c) Size of cuvette to be furnished (see 1.2).
(d) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

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6.3 This specification does not cover all types, classes, grades or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet requirements.

* 6.4 This specification covers the following cuvettes that are listed in the Federal Supply Catalog.

National Stock Number	Item Identification
6640-00-439-8420	CUVETTE, SPECTROPHOTOMETER, 10x75 mm., 12s.
6640-00-439-8440	CUVETTE, SPECTROPHOTOMETER, 19x150 mm., 12s.

* 6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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